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REVIEW

REVISED Weed science research achievements on maize in**Ethiopia: a review**

[version 4; peer review: 1 approved with reservations]

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Abstract

Recent research into assessing and combatting maize weeds was reviewed based on scientific and working reports from the year 2011 onwards in Ethiopia. This review paper covered two main topics: assessment of major weed types and their impacts, and research results of weed management in the maize cropping system of the country. More than 80 major weed floras dominated by broad leaved weeds were found infesting maize fields in different localities. By competing, toxifying and/or parasitizing maize plants, they reduce the crop germination, growth and productivity. Several experiments indicated that maize yield loss due to uncontrolled weed growth varied from 35 to 87.5%, indicating that weed management is a critical factor for maize production in the country. Hoeing, hand weeding and pre-emergence herbicides application have been the most frequently used weed control practices to minimize the competition effect of weeds. So far, there has been no effective and sustainable individual weed control method for maize fields. Thus, future research should focus on developing integrated weed management that effectively reduce weed infestation and associated impacts, and improve productivity of the crop considering socio-economic condition of a specific maize producing agro-ecology of the country.

Keywords

Ethiopia, maize, management, research achievements, weed

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REVISED Amendments from Version 3

Few fine tunings have been done in abstract, introduction, text body, conclusion and reference sections of the text. Based on reviewer and co-authors suggestions and comments, they were done to improve clarity and structure of subtitles, and sentences. In addition, the revisions have been made to explain well the symbols and abbreviations used in tables particularly in Tables 1, 2 and 3.

Any further responses from the reviewers can be found at the end of the article

Introduction

Maize (*Zea mays* L.) is an important cereal crop in Ethiopia because of its high productivity per unit area, suitability to major agro-ecologies and compatibility with many cropping systems. It is one of the high priority crops mainly cultivated during the rainy season for food, fodder and industrial raw material in the country. According to the [Central Statistical Agency of Ethiopia's, CSA \(2019\)](#) report, out of the total grain crop area (12.7 million hectares), maize took up 18.60% (2.4 M ha) standing next to tef (*Eragrostis tef*) with 24.17% (3.1 M ha) of occupation. According to the report, the production of maize contributed 30.08% (950,000 tones) of the grain production with a mean productivity of 4 t-ha⁻¹ in the country, which is by far less than its potential yield.

Major constraints to maize production in Ethiopia include both abiotic and biotic factors. The abiotic factor includes drought, nutrient deficiency, inadequate crop husbandry such as limited land preparation, planting dates and population densities adjustment, fertility management, and harvesting machinery. While the biotic factor includes low yielding varieties, weeds, diseases and insects. Among the biotic stresses, regular and noxious parasitic weeds (*Striga* species) are the most important limiting factors for maize production. They restrict the crop's yield by competing for nutrient, water, light and space, may vastly diminish its yield and cause complete failure of the plant due to parasitism and/or allelopathy. There are also indirect effects of weeds infestation on maize fields such as a reduction in land value, creation of inconvenience to agricultural operations and an increase in production costs. Furthermore, maize weeds act as hosts for plant pathogens and harbor insects; thus, aggravating the crop yield loss.

It has been observed that if no good control is carried out, weeds may cause a 35–87.5% reduction in maize yield ([Getahun et al., 2018](#); [Muluadam and Demisew, 2021](#)). Hence, weeds are critical factor in maize production system of Ethiopia, as a result their management is given the top priority compared to other maize pests. Hoeing, hand weeding and/or pre-emergence herbicides technology have been found the most frequently used weed control practices to minimize the competition effect of weeds in maize fields in the country.

Decade-based reports on the status of weed management research in maize in Ethiopia have been made starting from 1980's; [Rezene et al. \(1993\)](#), [Kassa et al. \(2002\)](#), and [Temesgen et al. \(2012\)](#). Efforts have been made to identify and characterize major weed species, assess crop losses due to the weeds interference, and identify effective management methods to reduce their impact. The objectives of this paper are to review the last decade (2011-2022) maize weeds research achievements, indicate research gaps and suggest future research directions.

Major weeds and their impacts on maize production system**Major weeds of maize**

In maize cropping systems, previous different farm management practices such as crop rotation, fertilizer application, herbicide usage and ploughing frequency are major factors affecting weed species types, distribution and infestation. On the other hand, weed flora are influenced by the current crop population density, sowing geometry, growth duration, growth rhythm, moisture period and fertility status ([Ali et al., 2016](#); [Muluadam and Demisew, 2021](#)). The moist and/or hot climate of the maize-growing belts of Ethiopia encourage rapid and abundant growth of weeds, consequently the crop fields are usually heavily infested with the weed plants. Various weed species considerably impede maize production, but their importance varies across maize producing agro-ecologies in the country.

Around 88 major weed floras under 23 plant families have been reported in maize across various localities ([Table 1](#)). Aboveground weed flora of 62 species in Abobo district of Gambella ([Dinberu et al., 2011](#)), 21 species in Bako ([Megersa and Fufa, 2017](#)), 15 species in Jimma ([JARC, 2021](#)) and six dominant weed species in north western Ethiopia ([Takele et al., 2022](#)) were recorded. According to the findings, majority of the encountered maize associated weed species were annual (71) as compared to perennial (17). Similarly, broad leaved weeds (66) outnumbered grass (18), sedge (2) and parasitic (2) weeds in the crop fields.

Table 1. Name and biological characteristics of major weeds of maize in Ethiopia, 2011–2022.

Family name	Name of species	Life cycle	Morphology
Acanthaceae	<i>Asystasia schimperi</i> T. Anders.	A	B
	<i>Hygrophila auriculata</i> Schumacher	A	B
	<i>Hypoestes forskalii</i> (Vahl.) R. Br.	A	B
	<i>Justicia heterocarpa</i> T. Anders.	A	B
Aizoaceae	<i>Trianthema pentandra</i> L.	A	B
Amaranthaceae	<i>Alternanthera sessilis</i> (L.) DC	P	B
	<i>Amaranthus caudatus</i> (L.)	A	B
	<i>Amaranthus dubius</i> Thell.	A	B
	<i>Amaranthus hybridus</i> L.	A	B
	<i>Cayusea abyssinica</i> (Fresen.) Fisch. & C.A.Mey.	A	B
	<i>Celosia argentea</i> L.	A	B
Asteraceae	<i>Acanthospermum hispidum</i> DC.	A	B
	<i>Ageratum conyzoides</i> L.	A	B
	<i>Bidens pachyloma</i> (Oliv. & Hiern) Cufod.	A	B
	<i>Bidens pilosa</i> L.	A	B
	<i>Cosmos bipinnatus</i> Cav.	A	B
	<i>Flaveria trinervia</i> (Spreng) C. Mohr.	A	B
	<i>Galinsoga parviflora</i> Cav.	A	B
	<i>Guizotia scabra</i> (Vis.) Chiov.	A	B
	<i>Gutenbergia rueppellii</i> Sch. Bip.	A	B
	<i>Medicago</i> spp.	A	B
	<i>Parthenium hyserophorus</i> L.	A	B
	<i>Periscaria nepalense</i> (Meisn.) H.Gross	A	B
	<i>Scorpiurus muricatus</i> L.	A	B
	<i>Sonchus oleraceus</i> L.	A	B
	<i>Xanthium abyssinicum</i> Wallroth	A	B
	<i>Xanthium spinosum</i> L.	A	B
Boraginaceae	<i>Heliotropium cinerascens</i> D.C.	A	B
	<i>Trichodesma zeylanicum</i> (L.) R. Br.	A	B
Caryophyllaceae	<i>Stellaria media</i> (L.) Vill.	A	B
Chenopodiaceae	<i>Chenopodium album</i> L.	A	B
	<i>Chenopodium murale</i> L.	A	B
Commelinaceae	<i>Commelina africana</i> L.	A	B
	<i>Commelina benghalensis</i> L.	P	B
	<i>Commelina latifolia</i> Hochst. ex A. Rich.	P	B
Convolvulaceae	<i>Ipomoea cordofana</i> Choisy in DC	A	B
	<i>Ipomoea fulvicaulis</i> (Hochst. ex Choisy.) Boiss	A	B
	<i>Ipomoea hochstetteri</i> House	A	B
Cucurbitaceae	<i>Cucumis ficifolius</i> A. Rich.	P	B
	<i>Momordica schimperiana</i> Naud.	P	B
Cyperaceae	<i>Cyperus distans</i> L. f.	P	Gy
	<i>Cyperus rotundus</i> L.	P	Gy
Euphorbiaceae	<i>Euphorbia geniculata</i> Orteg.	A	B

Table 1. Continued

Family name	Name of species	Life cycle	Morphology
Lamiaceae	<i>Leucas inflota</i> Beanth.	A	B
	<i>Leucas martinicensis</i> (Jacq.) Ait. f.	A	B
	<i>Ocimum basilicum</i> L.	A	B
	<i>Ocimum lamiifolium</i> Hochst. ex Benth.	A	B
Leguminaceae	<i>Euphorbia hirta</i> L.	A	B
	<i>Glycine wightii</i> (Wight and Am.) Verdc.	P	B
	<i>Indigofera amorphoides</i> Jaub. and Spatch.	A	B
	<i>Mimosa invisa</i> Mart. ex Colla	A	B
	<i>Senna didymobotrya</i> (Fresen.) Irwin & Barneby	P	B
	<i>Tephrosia pentaphylla</i> (Roxb.) G. Don	A	B
	<i>Trifolium steudneri</i> Schweinf.	A	B
Malvaceae	<i>Abutilon fruticosum</i> Guill & Perr.	P	B
	<i>Hibiscus aponeurus</i> Sprague & Hatch	A	B
	<i>Hibiscus micranthus</i> L. f.	A	B
	<i>Sida acuta</i> Burm. f.	P	B
	<i>Sida spinosa</i> sensu Cufod	P	B
Nyctaginaceae	<i>Commicarpus verticillatus</i> (Poir.) Stan.	P	B
Orobanchaceae	<i>Striga hermonthica</i> (Delile) Benth.	A	B
	<i>Striga asiatica</i> (L.) Kuntze	A	B
Phyllanthaceae	<i>Phyllanthus maderaspatensis</i> L.	P	B
Poaceae	<i>Brachiaria eruciformis</i> (Sm.) Griseb.	A	G
	<i>Cynodon dactylon</i> (L.) Pers.	A	G
	<i>Digitaria abyssinica</i> (Hochst. ex A.Rich.) Stapf	A	G
	<i>Digitaria ternata</i> (A. Rich.) Stapf.	A	G
	<i>Digitaria velutina</i> (Forssk.) P. Beauv.	A	G
	<i>Echinochloa colona</i> (L.) Link.	A	G
	<i>Eleusine jaegeri</i> Pilger	A	G
	<i>Eleusine indica</i> (L.) Gaertn.	A	G
	<i>Eragrost paniciformis</i> (A.Br.) Steud.	P	G
	<i>Eriochloa procera</i> (Retz.) C. Hubb.	A	G
	<i>Panicum atrosanguineum</i> A. Rich.	A	G
	<i>Panicum maximum</i> Jaquin	P	G
	<i>Pennisetum aphacelatum</i> (Nees) T. Durand & Schinz	P	G
	<i>Plantago lanceolata</i> L.	A	G
	<i>Setaria pumila</i> (Poir.) Roem. & Schult.	A	G
	<i>Snowdenia polystachya</i> (Fresen.) Pilg.	A	G
	<i>Sorghum verticilliflorum</i> (Steud.) Stapf.	A	G
	<i>Urochloa panicoides</i> P. Beauv.	A	G
Polygonaceae	<i>Oxygonum sinuatum</i> (Hochst. & Steud. ex Meisn.) Dammer	A	B
	<i>Polygonum</i> spp.	A	B
Solanaceae	<i>Nicandra physalodes</i> Scop.	A	B
	<i>Solanum indicum</i> L.	A	B
	<i>Solanum nigrum</i> L.	A	B
Tiliaceae	<i>Corchorus trilocularis</i> L.	A	B
Vitaceae	<i>Cissus quadrangularis</i> L.	A	B

Note: A = annual, P = perennial, B = broad leaf, G = grass, Gy = grassy.

Source: Dinberu et al. (2011), Megersa and Fufa (2017), JARC (2021), Takele et al. (2022).

Impact of weeds on maize

Weeds infestation immensely affects maize production and productivity in Ethiopia. Weeds force farmers to spend their effort (time and energy) on managing them; otherwise, they cause a considerable loss and this sadly is a common feature observed in sorghum and maize growing areas of the country (Muladad and Demisew, 2021).

Previous studies showed that weeds reduced maize yields by 35–87.5% depending on the crop cultivar types, density and sowing geometry; weed types, density and duration of competition; moisture period, fertility status, and tillage practice (Ali et al., 2016; Muladad and Demisew, 2021). Weeds interference on maize production system could be in different forms such as competition, allelopathy and/or parasitism.

Competition

Ethiopia, as one of the tropical arid and semi-arid countries confronts constraints due to a high temperature, erratic rainfall pattern, high rate of evapotranspiration, and fast and vigorous growth of weeds in crops (Lulseged et al., 2015; Muladad and Demisew, 2021). The rapid and vigorously growing weeds aggressively compete with crop plants for scarce soil resources like moisture and organic matter, and sunlight. In maize, weeds compete for growth factors and the crop yield losses due to competition have been estimated at different localities. Keeping maize weed-free throughout the cropping season is preferred to avoid the possible impacts, but considering the management cost it needs to determine the critical period of weed competition. A study by Muladad and Demisew (2021) showed that maize and sorghum crops are highly sensitive to weed competition, especially during their early growth stage.

Weed rating inversely impacted yield ($b = -0.5$; $p < 0.001$) where fields with the lowest weed rating had the highest yield (4.6 t ha^{-1}) than those with the highest rating (2.3 t ha^{-1}) (Lulseged et al., 2015). Accordingly, critical weed-free periods of 25–30 days after sowing (DAS) and optimum weed control periods of 30–50 days after emergence (DAE) have been developed and recommended for major weeds of maize, respectively. The critical timing of parthenium removal to avoid 5% and 10% maize grain yield loss was 8 and 17, and 13 and 23 DAE during year 2012 and 2013, respectively (Safdar et al., 2016). The relatively shorter critical period of parthenium competition in maize crop suggested its high allelopathic effect in addition to its competitive behavior.

Allelopathy

Allelochemicals such as phenolics, alkaloids, fatty acids, indoles, terpenes and others liberate from weeds through leaf leachates, decomposition of plant residues, volatilization and root exudates that affect crop productivity (Ali et al., 2016). Many studies have revealed the allelopathic effect of different weed species on maize seed germination, stand establishment, growth, yield and/or the physiology of the crop plants (Table 2). Allelopathic water extracts of *Plantago lanceolata*, *Anagallis arvensis*, *Medicago polymorpha*, *Phragmites australis*, *Ammi visnaga*, *Silybum marianum*, *Malcolmia africana*, *Emex spinosa*, *Calendula arvensis*, *Fumaria indica*, *Convolvulus arvensis* and *Rumex crispus* showed toxicity against maize, sunflower and wheat germination (Khan et al., 2012). Aqueous extracts of all plant parts (shoot, leaf, inflorescence, root and whole plant part) of *Parthenium hysterophorus* inhibited the germination and growth of maize, demonstrating the allelopathic potential of the weed plant in the crop field (Anteneh, 2015). Residues of leaves, stems, roots and whole plants of *Polygonum hydropiper*, *Amaranthus spinosus*, *Cyperus esculentus*, *Cyperus rotundus* and *Imperata cylindrical*, and their mixtures imposed an inhibitory effect on the emergence of maize seedlings (Ali et al., 2016).

Table 2. List of plant species with allelopathic effect on maize.

Species	Character
<i>Ageratum conyzoides</i> L.	B, I
<i>Amaranthus spinosus</i> L.	B
<i>Amaranthus viridis</i> L.	B
<i>Ammi visnaga</i> L.	B, M
<i>Anagallis arvensis</i> L.	B
<i>Calendula arvensis</i> (Vaill.) L.	B, M
<i>Calotropis procera</i> L.	B, I

Table 2. Continued

Species	Character
<i>Chenopodium album</i> L.	B
<i>Convolvulus arvensis</i> L.	B
<i>Cyperus esculentus</i> L.	Gy
<i>Cyperus rotundus</i> (L.) Myrabella	Gy
<i>Emex spinose</i> (Campd.) L.	B
<i>Eucalyptus camaldulensis</i> Dehnh	Tree
<i>Eupatorium adenophorum</i> Spreng.	B, M
<i>Fumaria indica</i> (Hauuskn.) Pugsley	B, M
<i>Imperata cylindrical</i> (L.) Raeuschel	G
<i>Lantana camara</i> L.	B, I
<i>Malcolmia Africana</i> (L.) R. Br.	B, I
<i>Medicago polymorpha</i> L.	B
<i>Parthenium hysterophorus</i> L.	B, I
<i>Phragmites australis</i> (Cav.) Trin. ex Steud.	G, I
<i>Plantago lanceolata</i> L.	G
<i>Polygonum hydropiper</i> L.	B
<i>Rumex crispus</i> L.	B
<i>Silybum marianum</i> (L.) Gaertn.	B, tistled

Note: B = broad leaf, G = grass, Gy = grassy, I = invasive, M = medicinal.

Source: Khan et al. (2012), Anteneh (2015), Ali et al. (2016), Awadallah and Eman (2017), Awadallah et al. (2017), Awadallah and Salwa (2019).

When added to soil at 5% and 10% concentrations, the inhibition of *Lantana camara*, *Ageratum conyzoides* and *Eupatorium adenophorum* residues on the emergence and shoot growth of wheat, rice and maize were reported by Ali et al. (2016). Eucalyptus (*Eucalyptus camaldulensis*) leaf extracts (20–100% of a 50 g/L solution) in laboratory and powders (0.5–3% w/w base) in greenhouse were found to have allelopathic effects on seed germination and seedling growth of the tested poaceous crops including maize (Awadallah and Eman, 2017). Similarly, pigweed (*Amaranthus viridis*) above-ground parts have allelopathic effects on seed germination and seedling growth of the tested poaceous crops including maize (Awadallah et al., 2017). According to Abiyu and Nagappan (2015); and Wakshum et al. (2018), *Lantana camara* leaf powder significantly inhibited seed germination, speed of germination, shoot and root length, stem thickness and biomass of wheat and maize. Moreover, *Calotropis procera* (sodom apple) leaves have allelopathic effects on seed germination and seedling growth of the tested poaceous crops including maize (Awadallah and Salwa, 2019).

Allelopathic potential has been also found in many crops like maize, rape seed, niger seed, sunflower, alfalfa, sorghum and wheat, and extracts of these crops contain allelochemicals that release from dead and decaying parts of the plant body (MuahmmadWaheed et al., 2013). If wisely planned, the application of allelopathy in cropping systems is quite effective in managing agricultural pests and improving the productivity of agricultural systems. The extract of maize root residues has been found effective as bio-herbicide in the control of *Bidens pilosa* (Modupe and Joshua, 2013). Several research findings have found that maize plant produces a number of secondary metabolites that inhibit weed plant growth at high concentrations. Having an allelopathic effect, the incorporation of *E. globulus* residues as green manure into soil could be a feasible practice to reduce the reliance on synthetic herbicides in maize-based cropping systems (Carolina et al., 2013).

Parasitism

Noxious root parasitic weeds like *Striga* spp. (witch-weeds) are widely spread in semi-arid regions of Africa including Ethiopia. The *Striga hermonthica* and *S. aspera* in northwestern and *S. asiatica* in the highlands of the eastern and southern parts of the country are threatening maize production. They are resulting in a complete crop loss under the worst conditions. According to Nagassa and Belay (2021), *Striga* spp. cause growth inhibition and yield losses of 20–100% in maize, sorghum, rice, pearl millet, finger millet and sugar cane. Prodigious seed production, prolonged viability of the seeds in soil and the subterranean nature of the early stages of parasitism in the host crops field make the control of the parasite difficult by conventional methods (Berhane, 2016).

In general, the occurrence of different weed species types and their infestation level vary from field to field mainly based on past farming history on a given field. The heavy infestation of weeds undoubtedly has considerable impacts on the germination, growth, production and productivity of crop plant. The different forms of weed interference on the crop plant indicate that weed management is a critical factor for maize production. Hoeing, hand weeding and pre-emergence herbicides application are the most frequently used weed control practices to minimize effect of weeds in maize fields of the country. However, so far none of them has been found effective and sustainable in controlling weeds in maize fields.

Weed management research results

Weeds are always present in the farming system and often the most significant pest of maize fields in Ethiopia. The level of weed pressure is important to the profitability of the crop production. The carryover effects of proper management practices might be a means of reducing the early weed load and then high crop production costs. Adequate weed management knowledge and technology availability are crucial for proper reduction of the impacts of weeds on the crop production. Maize cropping practices such as rotation, proper land preparation, growing of improved varieties, intercropping and soil fertility management are determinant for the effectiveness of the crop production. According to [Musa et al. \(2014\)](#), the impact of technical inefficiency of maize farmers is as high as 66%. It needs to design research and extension packages based on the model farmers' best practices.

[Tesfaye et al. \(2014\)](#) reported that proper weeds management in maize can increase the crop yield by up to 96%. Timely weeding makes the management effort effective than late weeding. It is labor constraint due to overlapping of activities during the cropping season that prevents timely weeding. When weeding delayed, costs of weeding increase and grain yield decline. Research is currently ongoing in the lowland tropics to determine the effect of reducing weed seed production by late weeding, zero tillage, and off-season management on weed pressure. Trials on the effect of farm practices such as crop rotations, intercropping, growing resistant or tolerant varieties, and maize-weed management are also being conducted ([WSRS, 2015](#)).

Cropping system: crop rotation and/or intercropping

Maize–haricot bean intercropping suppressed weeds, and was more productive and cost-effective than sole crop production in southern Ethiopia ([Workayehu and Wortmann, 2011](#)). Use of conservation agriculture-based maize–legume cropping systems was increased soil organic matter, total nitrogen, available phosphorus, and soil water infiltration rate by 37, 54, 12.3 and 327%, respectively, when compared to conventional tillage with maize monocropping over three years ([Yayeh et al., 2022](#)). By the conservation agriculture-based maize–haricot bean intercropping systems, weed infestation was reduced over the years and maize grain yield increased by 54% over the conventional tillage. Similarly, intercropping maize with forage legumes such as cowpea, silver-leaf and velvet bean has the potential to restore soil fertility and control weeds without reducing the crop grain yield ([Marilyn et al., 2022](#)).

Growing resistant and/or tolerant variety

Sowing an early emerging, fast-growing and competitive maize variety shall be made while cultivating the crop on weed-infested fields. Crop competition in maize may involve the use of competitive varieties that exhibit weed suppression potential. Weed-competitive varieties have a high leaf area index, and other leaf architecture elements that improve light interception by the crop, increasing the weeds shading capacity ([Blessing et al., 2016](#)). Another area that has been largely overlooked is the problem of parasitic weeds. *Striga* spp. inflict considerable losses on maize production, and no resistance has been found within the maize genome unlike for sorghum. Before releasing maize varieties in *striga* endemic areas, its genotypes must be screening for resistance and/or tolerance to the weed species.

Mechanical control; pre-planting tillage and/or post-planting inter-row cultivation

Farmer practices such as proper ploughing and inter-row cultivation are not given strong attention for weed management. The primary objectives of tillage are to prepare a smooth seedbed, alleviate soil compaction and reduce weeds. The clean seedbed is essential to reduce the weed seed bank in soil and reducing post-planting weed pressure. Lack of oxen energy for tillage, force farmers to practice usually minimum tillage that results in high grass weed infestation, thus allowing low labor for land preparation results with the high weed pressure in maize fields. According to [Bakhtiar et al. \(2011\)](#), the harvested maize grain yield was considerably higher by conventional tillage (2429 kg·ha⁻¹) than zero-tillage (2271 kg·ha⁻¹). The authors also indicated that wheel-hoe weeder is found more effective than manual and oxen labor particularly on properly tilled light soils that the implement can control 85% of the maize weed species.

Cultural control; hoeing, hand-pulling or mulching

Unlike many other pests, weeds can be controlled through hand weeding before they damage the crop if sufficient labor is available. Because of the limited resource base, the subsistence farming community relies on hand weeding for the control of weeds. As a result of the overlap of farm operations during the crop season; however, farmers either leave their farms

un-weeded or perform weeding late in the season. According to [Bakhtiar et al. \(2011\)](#), a considerably higher grain yield of 2863 kg·ha⁻¹ was recorded by practicing hand weeding compared to plastic or live plants mulching (2145 kg·ha⁻¹) or weedy check (1422 kg·ha⁻¹). However, besides weed pressure reduction, water conservation by mulching makes maize more resilient to the high rainfall variability occurring in some parts such as the Central Rift Valley of Ethiopia ([Sime et al., 2015](#)).

Good crop husbandry

The planting time, seeding pattern, rate and spacing; and fertilizer application played a significant role on maize yield ([Lulseged et al., 2015](#)). Lately planted maize experiences greater weed problems, nitrogen deficiency and water deficits usually result in lower grain yields. Then the maize yield reduces markedly and progressively as the time of planting delayed after the onset of the rain. The early-sown crop plants benefit by receiving full-season rainfall, suffering less weed competition and a high soil nitrate level at the beginning of the rains which subsequently declines due to leaching and a slower rate of humus decomposition. The actual utilization of nitrogen by the crop is highly affected by the weedy conditions of the field under which urea is applied.

Competition in maize may involve techniques such as reduced row spacing and increased planting density that exhibit weed suppressive potential. The row-sowing of maize saves seeds and fertilizer, makes weeding and other agronomic practices easier, and then provides high crop yields. Moreover, increasing the maize plant population from 30,000 to 60,000 plants per hectare was increased the crop yield from 2055 to 2412 kg·ha⁻¹ ([Bakhtiar et al., 2011](#)). The row planting of maize preconditions band-application of fertilizers that favors the crop growth more than inter-row weeds and also enhances productivity.

In Ethiopia, fertilizer application is a good input in improving the maize crop husbandry and farmers was mainly relied on commercial fertilizers to sustain livelihood security. Application of 150 kg·ha⁻¹ NPS (blended nitrogen, phosphorus and sulfur fertilizers) at sowing and 250 kg·ha⁻¹ urea splinted into two; half at knee height and the remained half before flowering stages are recommended for efficient weed control and maize growth improvement ([Takele et al., 2022](#)). Farmyard manure applications at the rate of 4 to 5 t·ha⁻¹ also suggested to favor the crop plants more than weeds. Presently, the unaffordable price of inorganic fertilizers has become a burning issue in the fertilizer-use habituated crop production system of the country, indicating the requirement to look for farmyard manure utilization.

Chemical control

Herbicides are preferred over manual weeding when they alleviate labor tension and/or increase yield by controlling weeds effectively. The effectiveness of various combinations and rates of post and pre-emergence herbicides were compared for their effect on weeds and productivity of maize in Ethiopia. Both broad leaf and grass weed control in maize with the herbicides have been suggested by many researchers and research centers of the country ([Firehywot et al., 2013](#); [Tesfaye et al., 2014](#); [Megersa and Fufa, 2017](#); [AmARC, 2020](#); [JARC, 2021](#)). The herbicide evaluation woks were compared and confirmed the reliability and promising potential of some products in effectively controlling weeds and enhancing yield of maize ([Table 3](#)).

Table 3. Herbicides type and rate verified for weeds control in maize fields, 2011–2021.

Herbicide trade name	Category	Rate
alazine 35/20 SE (alachlor 350 g/L + atrazine 200 g/L)	PrP	5 L·ha ⁻¹
atrazine (Aatraz)	PrE	2 kg·ha ⁻¹
atrazine + s-metolachlor (Primextra gold 720 SC)	PrE	2 kg·ha ⁻¹ (2)
BAKOMATE 670 SE	PrE	3 L·ha ⁻¹
Butrazine 48 SC	PrE	2.5 L·ha ⁻¹
Corn star	PoE	1.5 L·ha ⁻¹
Gesparim combi	PrE	3.5 kg·ha ⁻¹
herbazed (Glyphosate)	PrP	2 L·ha ⁻¹
Integrity EC 660 g/L	PrE	1 L·ha ⁻¹
Lasso + atrazine	PrE	5 L·ha ⁻¹
Lunar 537.5 SE	PrE	3 L·ha ⁻¹
LOGZO 337 EC	PoE	1.5 L·ha ⁻¹

Table 3. *Continued*

Herbicide trade name	Category	Rate
METRO GOLD 480 SC	PrE	1 L·ha ⁻¹
NICO-STAR 40 OD	PoE	1.5 L·ha ⁻¹
Nicosulfuron (Arrow 75 WDG)	PoE	0.09 kg·ha ⁻¹
pendimethalin (Stomp 330 EC)	PoE	1.50 kg·ha ⁻¹
PRIMASAVE	PrE	3 L·ha ⁻¹
round-up 36 SL (Glyphosate)	PrP	2 L·ha ⁻¹
s-metolachlor (Dual gold 960 Ec)	PrE	2 kg·ha ⁻¹
s-metolachlor + atrazine (Primagram gold 660 SC)	PrE	3 L·ha ⁻¹
Stellar™ star	PoE	1 L·ha ⁻¹
Surestart	PrE	3 L·ha ⁻¹
Tembo 80 OD	PoE	1.35 L·ha ⁻¹
Venus 500 SE	PrE	6 L·ha ⁻¹
weedknock (Glyphosate)	PrP	2 L·ha ⁻¹

Note: PrP = pre-planting, PrE = pre-emergence, PoE = post-emergence.

Source: Tesfaye et al. (2014), Megersa and Fufa (2017), Megersa et al. (2018, 2020), AmARC (2020), JARC (2021).

Firehywot *et al.* (2013) found that twice hand weeding, atrazine or s-metolachlor application at 2 L·ha⁻¹ resulted in higher weed control efficiency (> 94%) than other treatments in maize fields (Table 4). The highest maize grain yield of 4664 kg·ha⁻¹ was obtained with twice hand weeding and hoeing followed by atrazine 2 kg·ha⁻¹ (4534 kg·ha⁻¹) and s-metolachlor 2 kg·ha⁻¹ (4490 kg·ha⁻¹) application which were increased by 56, 55 and 54% over uninterrupted weed growth throughout the crop season. In addition, Megersa and Fufa (2017) reported that application of pre-emergence herbicides (Lunar 537.5 EC at 3 L·ha⁻¹, Venus 500 SE at 6 L·ha⁻¹ and s-metolachlor + atrazine at 3 L·ha⁻¹ with 200 L·ha⁻¹ water solution) considerably reduced weed infestation compared to weedy check plot there by boosted maize grain yield. JARC (2021) reported that pre-emergence herbicide Gesparim combi at a rate of 3.5 kg·ha⁻¹ and Lasso + atrazine at 5 L·ha⁻¹ supplemented with one hand weeding at 50–55 DAE kept the maize field weed-free throughout the season and gave the highest maize grain yield of 5,593 kg·ha⁻¹. In addition, the report indicated that s-metolachlor+ atrazine at 4 L·ha⁻¹ and alazine 35/20 SE at 5 L·ha⁻¹ were found effective in managing weeds in maize fields of the area.

Table 4. Effect of herbicides on above ground dry matter of weeds at different maize growth stages.

Herbicide	Weed dry matter (g/m ²)		
	25 DAS	60 DAS	Harvest time
atrazine 1.0 kg·ha ⁻¹	1.89 (3.1)	3.33 (10.7)	10.42 (108.1)
atrazine 1.5 kg·ha ⁻¹	1.37 (1.3)	2.44 (5.5)	8.68 (74.8)
atrazine 2.0 kg·ha ⁻¹	1.00 (0.5)	1.33 (1.3)	6.20 (38.4)
pendimethalin 1.0 kg·ha ⁻¹	1.52 (1.8)	4.83 (22.8)	19.65 (386.2)
pendimethalin 1.25 kg·ha ⁻¹	1.19 (0.9)	5.58 (30.7)	14.10 (198.4)
pendimethalin 1.5 kg·ha ⁻¹	1.09 (0.7)	5.13 (26.0)	14.54 (211.2)
s-metolachlor 1.0 kg·ha ⁻¹	1.55 (1.9)	4.22 (17.3)	11.78 (138.2)
s-metolachlor 1.5 kg·ha ⁻¹	1.26 (1.1)	3.71 (13.3)	10.61 (112.1)
s-metolachlor 2.0 kg·ha ⁻¹	1.05 (0.6)	2.41 (5.3)	6.69 (44.4)
once hand weeding (25 DAS)	2.99 (8.4)	3.58 (12.4)	14.57 (211.9)
twice hand weeding (25 & 45 DAS)	2.96 (8.3)	0.95 (0.4)	5.85 (33.7)
weedy check	3.00 (8.5)	11.25 (126.0)	26.81 (718.5)
LSD (0.05)	0.18	0.44	0.77
CV (%)	6.28	6.33	3.63

Source: Firehywot *et al.* (2013).

Table 5. Interaction effect of weed management practices and inter-row spacing on grain yield (kg·ha⁻¹) of maize during 2018 cropping season at Bako.

Weed management practices	Inter-row spacing (cm)			
	65	70	75	80
Weedy check	3014i	2771ij	2597ij	2237j
S+A at 1 L·ha ⁻¹	7517fg	7024g	6347h	6191h
S+A at 2 L·ha ⁻¹	9634bc	8687de	7707f	7393fg
S+A at 3 L·ha ⁻¹	9945ab	9237cd	8452e	7521fg
Hand weeding	10492a	9922ab	8790de	7705f
LSD (5%) WMP * IRS = 565.0, CV (%) = 4.8				

Note: Means followed by the same letter within column are not significantly different at 5% P, S+A = s-metolachlor + atrazine, Hand weeding = hand pulling and hoeing at 25 and 45 DAS, LSD = least significant difference and CV = Coefficient of variation.

Source: [Megersa et al. \(2020\)](#).

Integrated weed management strategy

In maize fields, integrated management strategy aimed at combining the economically feasible, environmentally safe and socially acceptable weed control options ([Takele et al., 2022](#)). This could be the incorporation of two or more weed control methods such as prevention, cultural, biological and/or chemical control to reduce their interference while maintaining acceptable yields of the crop. If various weed management components are integrated in a systematic manner, a significant advances in weed management technology can be achieved.

[Gebisa and Gressel \(2007\)](#) reported that a combination that included resistant varieties, fertilizer and tied-ridge gave significantly higher sorghum yield followed by one that combined local varieties with fertilizer and tied-ridge in north Wollo. In semiarid areas of the Central Rift Valley, conservation agriculture by covering soil with crop residues reduced soil water evaporation, erosion as well as degradation; water run-off and weed problems, and then improved soil fertility, water infiltration, soil biological activity, soil water availability and maize-bean productivity ([Liben et al., 2017](#)).

Similarly, glyphosates such as herbazed, round-up 36 Sl and weedknock at 2 L·ha⁻¹ rate mixed in 200 L·ha⁻¹ water on 15–18 days before planting application supplemented by hand weeding on 40 DAS provided efficient weed control and then considerably boosted maize grain yield at Bako ([Megersa et al., 2018](#)). Twice-hand weeding and a plant density of 53,333 plants per hectare were found to be suitable practices for attaining optimum grain yield for the hybrid maize BH-546 in Assosa district ([Getahun et al., 2018](#)).

The application of s-metolachlor + atrazine at 2 L·ha⁻¹ in combination with 65 cm inter-rows spacing (IRS) found as the best option for effective weed management in maize at Bako ([Megersa et al., 2020](#)). Furthermore, integration of the narrower IRS with hand pooling plus hoeing improved weed control efficiency there by enhanced maize grain yield ([Table 5](#)).

The increasing incidence of striga has been attributed to poor soil fertility, low soil moisture and the intensification of land use through continuous cereal cultivation. Thus, potentially successful approaches developed to control this weed include rotation with trap crops or legumes, applying pre-emergence herbicides, growing resistant or tolerant varieties, sowing clean seeds, intercropping with legumes, soil amendments with fertilizer or manure, water conservation, use of push-pull technology and effective microbes (*Fusarium oxysporum* and arbuscular mycorrhizal fungi) as biological control agent ([Nagassa and Belay, 2021](#)). Hence, to successfully reduce infestation pressure and the impact of the parasitic weed, an integrated management strategy needs to be developed considering the ecological and socio-economic conditions of the major maize growing areas of the country.

Conclusions

The potential yield of maize varieties is very interesting in Ethiopia. There is a large gap between the potential and actual yields of the crop. Weeds are one of the major biotic factors contributing to the low maize yield in the country. Effective management is substantive in reducing the broad interference of various weed species as well as their negative impact on the crop productivity. Few individual weed control methods identified as part of a package for the improved varieties. Strengthened research activity is needed to ensure that maize yield losses from stresses are minimized through sustainable and strong weed management technology against maize weeds; regular, parasitic and/or allelopathic weeds. Current weed science research information reviewed in this paper is very rare despite the expansion of maize production areas and

associated weed infestation problems. Closing the technology gap requires the diffusion of advanced weed management technology against maize weeds. There is a pressing need particularly by smallholder farmers to adopt integrated weed management strategies at different agro-ecologies of the country.

Research gaps and future directions

Weed species distribution and interference is dynamic causing consistent maize yield reduction in Ethiopia. However, there is no strong weed science research activity, and then latest information, knowledge and technology that can assist to adequately manage maize weeds in the country. Parasitic and allelopathic effect of weeds are often overlooked, yet they are likely important as potential stress on maize plant physiological functions and productivity. So far there are few individual weed control methods that ineffectively adopted by maize growers, and there is lack of dependable integrated weed management strategy for every maize agroecology of the country.

The dynamic nature of weed species' demands intensive weed science research endeavor in Ethiopia. In the absence of the latest information, knowledge and technology on appropriate weed management measures, it is hardly possible to maximize the grain yield of maize. It could be good to frame future avenues of research that could help to move studies of competition, parasitism and allelopathy into the broader ecological context. In general, the weed management research at representative major maize growing locations shall be strengthened within the context of developing an integrated weed management approach that considers the social, economic and agro-ecological conditions of a specific locality. Moreover, it needs to upgrade the knowledge and capacity of smallholder farmers by promoting the improved weed management technologies.

Data availability

All data supporting the results are included in the article and no extra source data are necessary.

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The authors highlighted the importance of maize in Ethiopian agriculture, highlighted the main weed species affecting maize crops, and explored the consequences of weed infestations on maize production. This article can be accepted, Provided you add relevant references for each piece of information given (there are complete sentences without references) and add a summary diagram revealing the importance of this study and their significant contribution. In addition, adding a table summarizing studies conducted in Ethiopia on weed management in maize could improve the quality of the review.

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Agroecological Weed Management

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Version 3

Reviewer Report 06 November 2023

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The article entitled "Weed Science Research Achievements on Maize in Ethiopia: A Review" authored by Negewo et al., offers a comprehensive assessment of weed management in maize production in Ethiopia. The authors emphasized the significance of maize in Ethiopian agriculture, highlighted the primary weed species affecting maize crops, and explored the consequences of these weed infestations on maize production. Furthermore, they delved into various weed management practices and evaluated their effectiveness in controlling weed proliferation. This study provides informative insights into the critical facets of agriculture in Ethiopia.

Some general and specific assessments and recommendations for the author are as follows:

Clarity and Structure: This article maintains a generally sound structure with a lucid introduction and well-defined section headings. However, there is room for improvement in terms of clarity and structure. One suggestion is to introduce more concise and structured summaries or introductions at the outset of each section. This will enable readers to swiftly grasp the key points without requiring a full reading of the section.

Additional Data and Analysis: This article provides a good introduction to the main weed species and their effects on maize. However, it would be useful to include more data and conduct a more detailed analysis. Adding statistics or case studies to support claims would enhance an article's informative value and credibility.

Address Research Gaps and Future Directions: While this article briefly touches upon research gaps and future directions, it is advisable to expound on these aspects. A comprehensive elaboration of the research required to bridge these gaps and its practical implementation would be beneficial.

Visual Aids: Enhancing the article with visual aids such as charts or graphs to depict data and trends related to weed infestations and management practices could make the content more engaging and comprehensible.

Language and Editing: The manuscript would benefit from additional proofreading and editing. Addressing minor grammatical issues and refining awkward sentence structures will enhance readability.

Symbol Definitions: The author should consider including a legend or explanation for the symbols and abbreviations used in the tables, particularly in Tables 2 and 3. This will contribute to the clarity of the data presentation.

Citations and References: It is essential to ensure that all statements and information presented in an article are substantiated by relevant and recent references. Comprehensive and accurate citations are fundamental for establishing the credibility of information.

Bibliographic Research Methodology: A brief description of the bibliographic research methodology employed, including the keywords used and databases targeted, should be included. This will aid readers in understanding how pertinent literature has been gathered.

Comprehensive Citation of Studies: The author should verify that the article thoroughly cites and references all pertinent studies on weed management in corn crops in Ethiopia to ensure the comprehensiveness and academic rigor of the review.

Inclusion of a Summary Table: To align with the article's title, the author is encouraged to incorporate a summary table that offers an overview of diverse studies conducted in Ethiopia concerning weed management in corn. This table should encompass details, such as research objectives, primary findings, key weed species, crop type, study areas, and proper citations for each study.

Recommendations for Agriculture: An imperative addition would be a section within the article outlining recommendations for agricultural practices, based on the findings of the review. Such recommendations would underscore the practical relevance of this review and elucidate its potential contribution to the development of agriculture in Ethiopia.

Incorporating these suggestions will not only enhance the article's quality and completeness but also render it more informative and valuable to its readers.

We hope that the authors find this review report to be helpful.

Best regards,
Dr. Abdellatif Boutagayout

Is the topic of the review discussed comprehensively in the context of the current literature?

Yes

Are all factual statements correct and adequately supported by citations?

Partly

Is the review written in accessible language?

Partly

Are the conclusions drawn appropriate in the context of the current research literature?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Agroecological Weed Management

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

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